DERIVED COMPLETENESS IS EQUIVALENT TO BEING THE DERIVED LIMIT OF THE TENSOR PRODUCTS WITH THE KOSZUL SYSTEM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 26341-26350 (canonical tag 091Z) and the counted proof is the proof environment at source lines 26352-26385, which immediately follows it and is its own proof.

Editorial note. The counted statement is formulated inside the situation of the section; that situation environment is delivered as the context span of this card, with its boundaries taken exactly at the opening and closing lines of the environment, so the objects named in the statement can be read and the reference inside it resolves in the compiled document.

Editorial note. The three stubs of this card cover the functoriality lemma cited inside the situation, the statement giving one implication and the exactness statement; their kinds are the type words appearing in the labels themselves, following the lane convention.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- The derived limit of the tensor products with the Koszul complexes is derived complete with respect to the ideal, which gives one implication. invoked at proof lines 26353-26355. Disposes of the implication from the second condition to the first.
- The construction sending an object to the derived limit of its tensor products with a system is exact in the object variable, so a filtration reduces the statement to its graded pieces. invoked at proof lines 26364-26368. Reduces the converse to the vanishing of the derived limits of the graded pieces of the filtration.
- The graded pieces of the stupid filtration of the Koszul complex are direct sums of copies of the quotient by the ideal up to shifts, whose derived limits of tensor products are computed by the torsion functors. used at proof lines 26369-26384. Identifies the derived limits in question with direct sums of torsion functors, which vanish by the assumption of derived completeness.

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Source reference. Original lemma 0624: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6612–6612. The reference occurs inside the situation environment delivered as the context span, where it produces the inverse system of Koszul complexes; it is a pointer needed only to define the objects of the statement.

Source reference. Original lemma 091Y: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26329–26329. The reference is cited at proof lines 26353-26355 for one implication; the converse, which is the substance of the proof, is inside the counted span.

Source reference. Original lemma 091K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23915–23915. The reference is invoked at proof lines 26364-26366 to reduce the statement to the graded pieces of the filtration; the explicit computation of those graded pieces is inside the counted span.

Source reference. Original section 0118: sources/stacks-chow/tags.txt, source lines 1354–1354. The reference points to the framework of stupid truncations used to build the filtration, cited at proof lines 26357-26358; it is a pointer to the construction and the use made of the filtration is inside the counted span.

Situation 0.2. Let A be a ring. Let $I = (f_1, \ldots, f_r) \subset A$. Let $K_n^\bullet = K_\bullet(A, f_1^n, \ldots, f_r^n)$ be the Koszul complex on $f_1^n, \ldots, f_r^n$ viewed as a cochain complex in degrees $-r, -r+1, \ldots, 0$. Using the functoriality of Lemma 0624 we obtain an inverse system

$$\ldots \to K_3^\bullet \to K_2^\bullet \to K_1^\bullet$$

compatible with the inverse system $H^0(K_n^\bullet) = A/(f_1^n, \ldots, f_r^n)$ and compatible with the maps $A \to K_n^\bullet$.

Lemma 0.3. *In Situation 0.2. Let $K \in D(A)$. The following are equivalent*

1. *K is derived complete with respect to I, and*
2. *the canonical map $K \to R\lim(K \otimes_A^{\mathbf{L}} K_n^\bullet)$ is an isomorphism of $D(A)$.*

Proof. If (2) holds, then K is derived complete with respect to I by Lemma 091Y. Conversely, assume that K is derived complete with respect to I. Consider the filtrations

$$K_n^\bullet \supset \sigma_{\geq -r+1} K_n^\bullet \supset \sigma_{\geq -r+2} K_n^\bullet \supset \ldots \supset \sigma_{\geq -1} K_n^\bullet \supset \sigma_{\geq 0} K_n^\bullet = A$$

by stupid truncations (Homology, Section 0118). Because the construction $R\lim(K \otimes E)$ is exact in the second variable (Lemma 091K) we see that it suffices to show

$$R\lim \left(K \otimes_A^{\mathbf{L}} (\sigma_{\geq p} K_n^\bullet / \sigma_{\geq p+1} K_n^\bullet)\right) = 0$$

for $p < 0$. The explicit description of the Koszul complexes above shows that

$$R\lim \left(K \otimes_A^{\mathbf{L}} (\sigma_{\geq p} K_n^\bullet / \sigma_{\geq p+1} K_n^\bullet)\right) = \bigoplus\nolimits_{i_1, \ldots, i_{-p}} T(K, f_{i_1} \ldots f_{i_{-p}})$$

which is zero for $p < 0$ by assumption on K. □